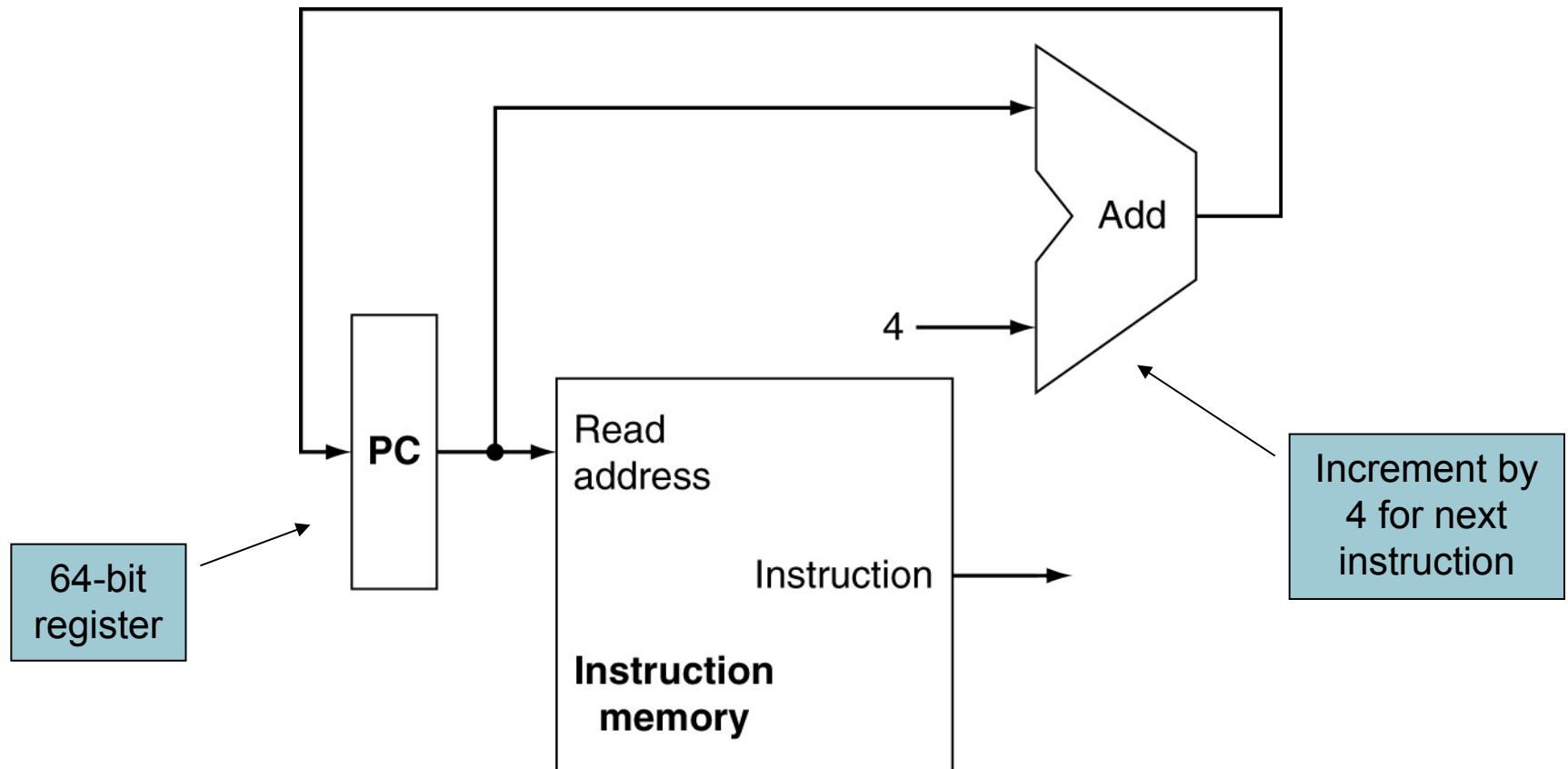


Chapter 4

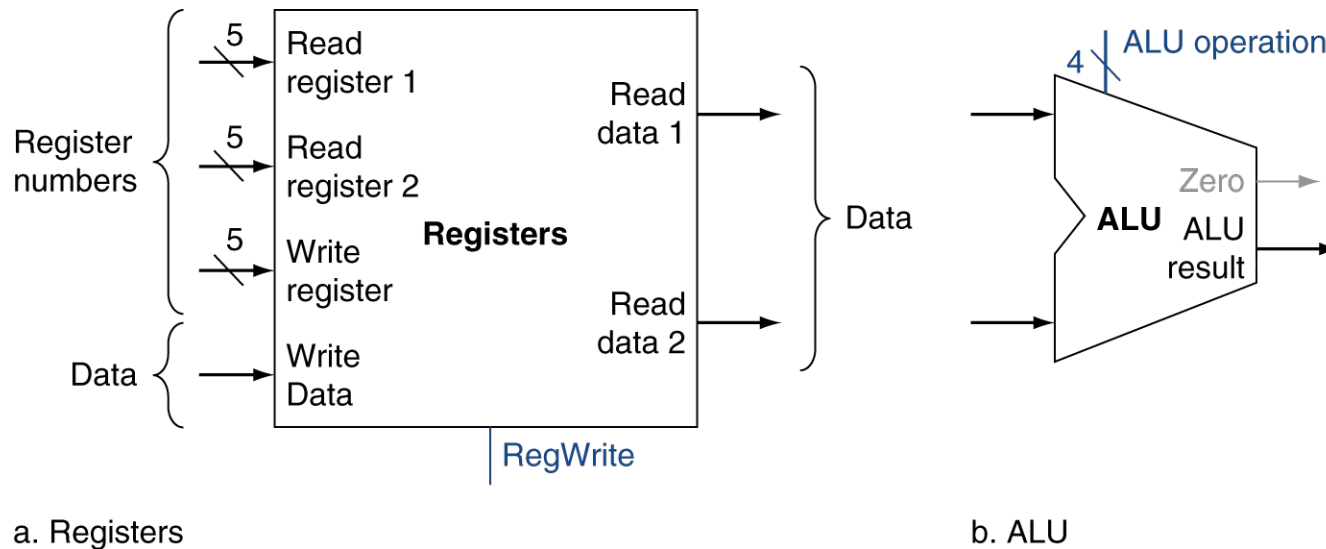
The Processor

Revision: Instruction Fetch



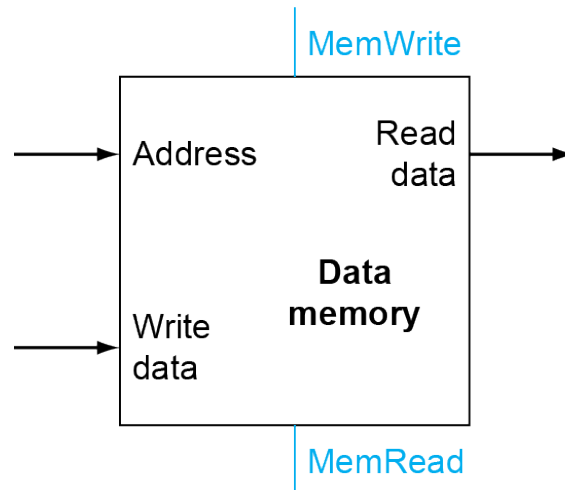
Revision: R-Format

- Read two register operands
- Perform arithmetic/logical operation
- Write register result

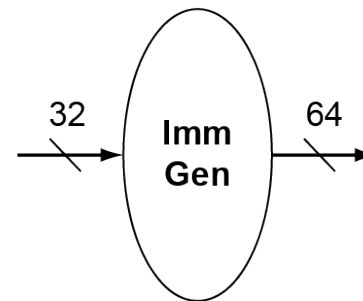


Revision: Mem Instructions

- Read register operands
- Calculate address using 12-bit offset
 - Use ALU, but sign-extend offset
- Load: Read memory and update register
- Store: Write register value to memory



a. Data memory unit

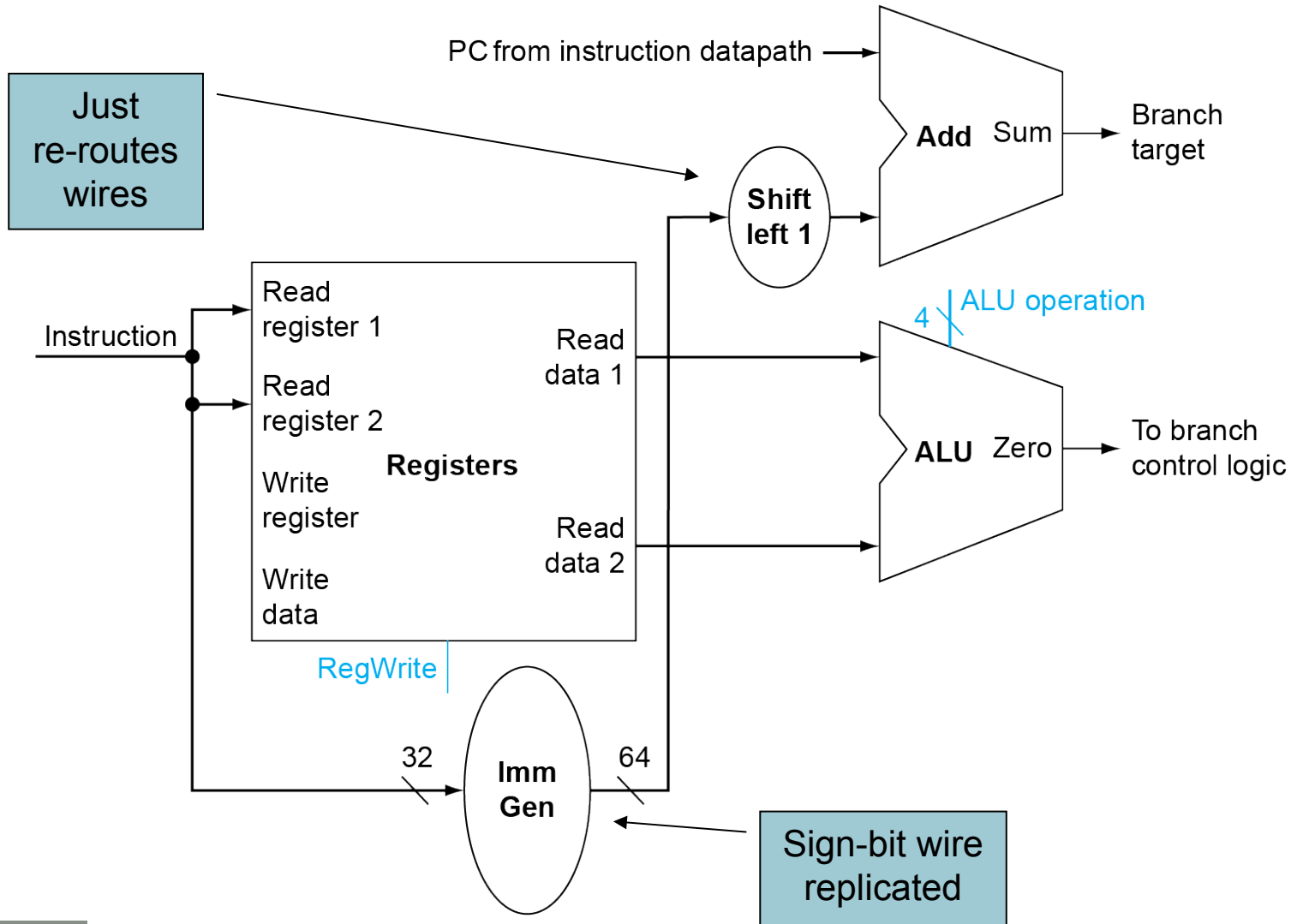


b. Immediate generation unit

Branch Instructions

- Read register operands
- Compare operands
 - Use ALU, subtract and check Zero output
- Calculate target address
 - Sign-extend displacement
 - Shift left 1 place (halfword displacement)
 - Add to PC value

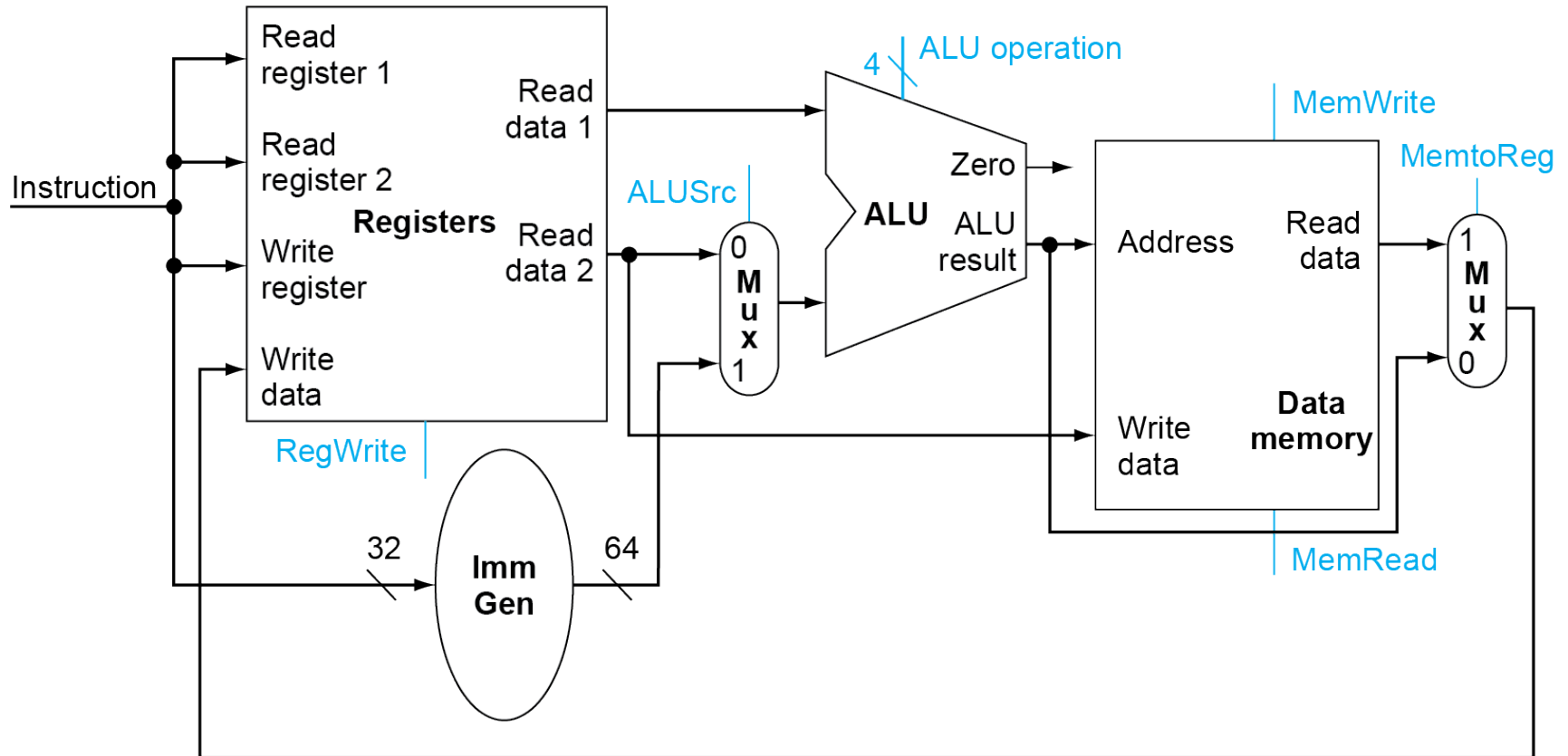
Branch Instructions



Composing the Elements

- First-cut data path does an instruction in one clock cycle
 - Each datapath element can only do one function at a time
 - Hence, we need separate instruction and data memories
- Use multiplexers where alternate data sources are used for different instructions

R-Type/Load/Store Datapath



Full Datapath

